

AGCD: Auditing Failure-Driven Policy Improvement in World–Action Models

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Abstract—Action-conditioned manipulation predictors can score candidate actions for the wrong reason. Perturbing expert actions and retaining terminal outcomes injects an *action-amplitude shortcut*: magnitude encodes success while geometry is ignored. We release AGCD, an intervention test that audits this leak, collects action-matched/geometry-varied counterfactuals, and verifies the same predictor by closed-loop selection. Action-L2 reaches AUROC 0.90 on the victim recipe but collapses to 0.50 under paired geometry. The same MiniWAM trained on perturbation versus paired data yields paired AUROC 0.50 vs 0.99 and closed-loop success 0.29 vs 0.96 (gap 0.67, 95% CI [0.52, 0.79]); FACT/RGB/DINOv2 variants retain gaps of 0.62–0.65. Distilling evaluator feedback into the same policy gives success 0.296 vs 0.976 for confounded vs paired feedback, with no evaluator query at test time. Physics-based PyBullet push and carry-release preserve gains of 0.722 and 0.651; in the latter, confounded feedback avoids drops yet sends 65.1% of trials to wrong targets. With frozen DINOv2 input and an independently BC-initialized policy, paired-feedback success remains 0.84–0.96 under noise, blur, crop, occlusion, and color shifts, improving over no update by 0.24–0.30. Fix the feedback contract before using failures to improve a policy. Generators, baselines, and audit gates: <https://anonymous.4open.science/r/icra-8056/>.

Index Terms—world model, counterfactual, shortcut learning, diagnostic suite, manipulation

I. INTRODUCTION

When a world-action model (WAM) ranks candidate actions by predicted progress or risk [1], [2], [4], the ranking is useful only if it reflects *action–geometry interaction*. If the score is a disguised function of action magnitude, selective execution is internally consistent and physically empty.

a) The closed-loop question.: AGCD asks whether holding the action fixed while changing visible geometry changes the predictor’s decision in the same direction as the rollout. Thus the paper follows one chain: confounded collection, ranking failure, paired collection, and ranking recovery on unseen layouts.

That failure mode is not hypothetical as a compositional risk. Recovery pipelines perturb demonstrations [6], while benchmark collectors expose terminal task outcomes [7]; outcome heads separately train on progress/value targets [4]. Connecting these ingredients—supervising an outcome head with expert-perturbation success—creates an *action-amplitude shortcut*: large perturbations fail, small ones succeed, so $\|a - a_{\text{nom}}\|$ predicts y without consulting geometry.

b) Positioning.: We study a *compositional* risk, not a smear of named checkpoints. FLARE’s perturbations are for recovery; FACT’s value heads are for failure-aware futures;

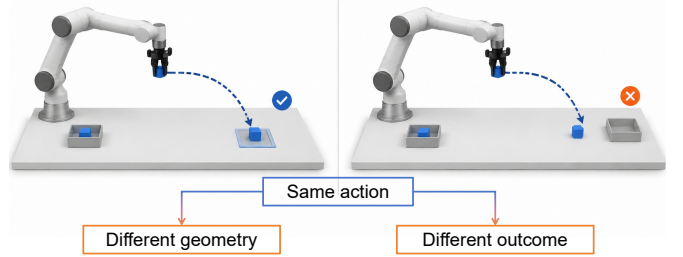


Fig. 1. **Motivating observation.** The same action yields different outcomes when target geometry changes. AGCD tests whether a predictor uses this interaction rather than action amplitude.

neither paper advocates the dangerous wiring (Table I). The risk appears when “perturb demos → keep success → train value” becomes the default next step—a wiring that is easy to invent once both ingredients exist in the ecosystem. AGCD makes that step falsifiable *before* it scales: detect the leak, ship an intentional victim, and show that changing the *collection protocol* (not the loss) restores decision quality.

c) Concrete failure.: On a fixed pick-place layout, replay the expert with joint noise $\sigma \in \{0.02, 0.04, 0.06, 0.08\}$. Large- σ episodes drop; small- σ episodes succeed. An outcome head can ignore seats and still post high AUROC by reading amplitude. Now hold the *same* action and move the target: amplitude is constant, yet success flips with geometry. Any model that learned the first cue fails the second—and closed-loop $\arg \max$ over matched directions collapses onto the min-L2 heuristic. The diagnostic question is whether changing only how labels are collected restores geometry-aware ranking.

d) Contributions.:

- 1) **Intervention test**: an A/B audit identifies whether an outcome predictor can solve the recipe from action amplitude alone.
- 2) **Paired protocol**: an action-matched factorial collector removes that cue while preserving action–geometry interaction.
- 3) **Learning closure**: the same heads and candidates are evaluated offline, in closed-loop selection, and through evaluator-guided policy updates whose held-out execution no longer queries the evaluator.

II. RELATED WORK

a) Action-conditioned prediction.: Action-conditioned video prediction established the basic premise that a robot

TABLE I
RECIPE AUDIT OF THE DANGEROUS CHAIN (PERTURB EXPERT $\rightarrow$ KEEP SUCCESS $\rightarrow$ TRAIN OUTCOME/VALUE). CELLS SUMMARIZE THE CITED WORK; THE GENERIC COLLECTOR ROW IS A POSSIBLE COMPOSITION, NOT A CLAIM THAT ANY RELEASED SYSTEM TRAINS THIS FULL CHAIN.

Work	Expert pert.?	Success as label?	Outcome/value head?	Full chain?
FLARE [6]	yes (Retry bridges)	no (policy target)	no	no
FACT [4]	no (policy fails)	yes (progress/value)	yes	no
CoCo [3]	CF for consistency	n/a	no	no
ACT-Bench [12]	eval only	no	metrics	no
Validate the Dream [13]	n/a	n/a	accreditation	no
Generic RL-Bench-style collector	yes (possible)	yes (terminal)	optional next step	<i>not inherent</i>
AGCD victim (ours)	yes	yes	yes	yes (intentional)

can learn physical consequences directly from interaction videos [23]; visual foresight then connected those predictions to model-predictive control on a real robot [24]. Latent world-model planning provides the corresponding compact-state formulation [25]. WAMs couple action generation with future prediction or consequence scoring [4], [5], [8]–[11]. PointWorld predicts full-scene 3-D point flow conditioned on robot point flow [17]; Diffusion Policy illustrates why modern policies naturally emit multi-step candidate chunks rather than isolated controls [18]. These developments make counterfactual ranking useful, but also make a scalar property of the candidate chunk available to every downstream head. ACT-Bench and Validate the Dream test action fidelity and accreditation [12], [13]; CoCo targets visual inertia with a consistency objective [3]. AGCD instead audits whether the *outcome labels themselves* reveal the answer through action amplitude.

b) Failure data and causal confusion.: Table I audits a compositional risk rather than naming a guilty checkpoint. FLARE perturbs demonstrations for recovery [6], but does not train an outcome head on the success of those perturbations. FACT trains progress/value on policy failures [4], not on expert-action perturbations. Classical DAgger changes the visited-state distribution by online aggregation [19]; causal confusion in imitation learning shows that predictive features can still identify the wrong causal mechanism under intervention [20]. Our failure is a concrete label-construction instance: perturbation scale jointly changes both the action and its probability of success. Shortcut learning and invariant risk minimization motivate environment interventions, but do not specify the manipulation data crossing needed here [14], [15]. Motus2 routes failed and suboptimal interactions to simulation/value learning and feeds value estimates back into planning and policy updates [5]. AGCD asks the complementary upstream question: whether the failure recipe makes that feedback identifiable from action–geometry interaction before it is allowed to update the policy.

c) Robot data and decision confidence.: RoboNet, BridgeData V2, DROID, and Open X-Embodiment provide broad robot experience across platforms and scenes [16], [21], [22], [26]; robomimic systematizes offline demonstration evaluation [27], while LIBERO and CALVIN stress task transfer and long-horizon control [28], [29]. Scale and

diversity do not by themselves supply outcome-balanced, same-action/different- geometry interventions; Sec. IV-E tests this boundary rather than treating demonstrations as a failure benchmark. Calibration and selective prediction make a valid score operational [30], [31]; conformal risk control can additionally bound a chosen monotone loss on held-out calibration data [32]. However, temperature scaling preserves ranking and rejection cannot repair a target whose shortcut is already encoded in the labels. AGCD is therefore upstream of calibration: first establish that geometry changes the ranking, then calibrate or abstain.

III. THE AGCD DIAGNOSTIC

A. Threat model

Let a be an action chunk, g scene geometry, and $y \in \{0, 1\}$ terminal success. A consequence predictor $f(a, g)$ is useful for selective execution only if its ranking depends on the interaction (a, g) , not on a scalar summary of a alone. Expert-perturbation CF draws $\tilde{a} = a_{\text{nom}} + \epsilon$ on fixed g and labels by rollout success. Write $s = \|\tilde{a} - a_{\text{nom}}\|$ (or mean per-step L2). If $\text{AUROC}(s \rightarrow y)$ is high on the training recipe while $\text{AUROC}(s \rightarrow y) \approx 0.5$ under paired geometry, then labels are not pure action–geometry interactions: $y \not\perp g \mid s$ fails in the direction that matters for ranking. Empirical risk minimization may ignore g and still look strong in-domain.

Noise for coverage is fine. Using noise-induced success as outcome supervision *without* amplitude control is the threat (Fig. 2). Paired-geometry collection is the complementary intervention: hold a fixed (or scalar-matched) and vary $\{g_k\}$ so that s carries no label information.

B. Identification by factorial crossing

The minimal paired unit is not a positive/negative trajectory pair; it is a crossing of action identity and geometry. Let $A = \{a_i\}_{i=1}^m$ be candidate directions and $G = \{g_j\}_{j=1}^n$ be layouts. We execute every cell (a_i, g_j) and obtain the rollout label

$$y_{ij} = \mathbb{I}[\text{task succeeds after executing } a_i \text{ in } g_j]. \quad (1)$$

The design is valid only when neither row nor column identity determines the label: each action succeeds and fails across layouts, and each layout succeeds and fails across actions. In the three-way controlled task this yields a permuted

AGCD: Action-Geometry Consequence Diagnostic

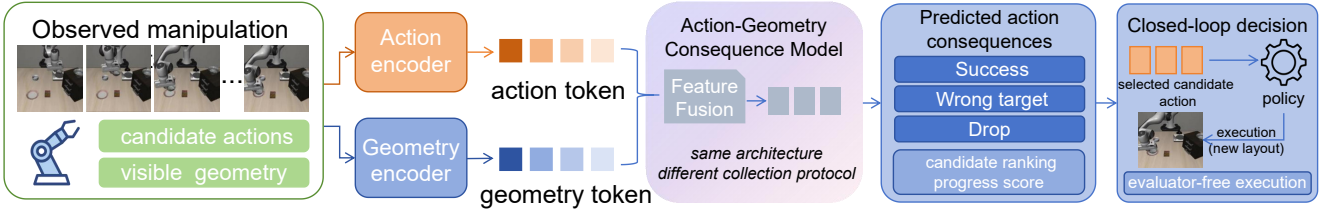


Fig. 2. **AGCD method overview.** Observed manipulation states, visible geometry, and candidate actions are encoded and fused to predict action consequences (success, wrong target, or drop), rank candidates, and make a closed-loop execute-or-reject decision.

diagonal pattern: matching direction and target succeeds; the other cells become wrong-target or drop. This is stronger than swapping one target under one fixed action, where geometry identity alone would reveal the label.

a) Why the amplitude probe must fail on valid pairs.:

Let $m(a)$ collect all scalar action summaries accessible to a shortcut: mean/max step L2, path length, peak speed, and gripper-transition count. If the crossing is balanced such that $p(m(a) | y=1) = p(m(a) | y=0)$, any score measurable only from $m(a)$ has AUROC 0.5 in expectation. This is not a claim that action is irrelevant; action *direction* is essential through its interaction with geometry. It says only that a marginal scalar cannot rank the outcomes. AGCD tests this condition empirically rather than assuming that equal command length implies equal robot motion.

b) *What is and is not identified.*: The factorial protocol identifies whether a predictor uses information beyond the audited action scalars on the support of the crossed layouts. It does not identify the causal effect of arbitrary actions, guarantee extrapolation beyond that support, or make simulator labels real. Consequently, we require layout-disjoint evaluation, action-only and geometry-only controls, and a closed-loop ranking test. A model passes only when the joint input improves over both marginals and selects the correct action on unseen layouts.

C. Three probes

Probe A (detect). Build a confounded set (fixed layout, noise ladder) and a paired set (fixed amplitude, varied geometry). Report $\text{AUROC}(s \rightarrow y)$ on both, plus transfer of an amplitude head trained on confounded and tested on paired.

Probe B (remove). Collect paired-geometry CF: hold a element-identical (or scalar-matched across directions) while varying $\{g_k\}$. Amplitude carries no label information; success requires alignment of a with g .

Probe C (decide). Train the *same* action-conditioned head on each recipe; evaluate both on a shared paired holdout and closed-loop arg max over three matched directions. We report a small MiniWAM (value + short future) and a *FACT*-style temporal Transformer: action-conditioned progress sequence + terminal value + short future (2.3M params), matching *FACT*’s scoring interface rather than its full video backbone. This is intentional. Reproducing *FACT*’s pixel

world model would conflate video modeling capacity with label leakage; our question is whether a larger progress/value head still falls for the amplitude shortcut when only the collection recipe changes. Architecture is not the claim—the claim is the protocol.

D. Collection contract

For every decision state, the collector writes a row containing: episode and layout identifiers; the full action chunk; the action-summary vector $m(a)$; target and object geometry; terminal object pose; contact-mode outcome; and success. A paired group is accepted only if (i) all prescribed directions are present, (ii) action horizons and gripper schedules agree, (iii) scalar spread is below the configured tolerance, and (iv) at least two outcomes occur. These checks happen before fitting any predictor. Groups that fail are diagnostics, not training examples.

Confounded and paired recipes use separate collectors but the same feature contract, architecture, and layout-level split. Geometry-only and action-only baselines reuse those indices, so the joint gain cannot come from an easier split or extra examples.

a) *Reset-free approximation.*: Without exact resets, we retrieve blocks within skill, phase, gripper state, and a pre-action visual-state neighborhood; impose calipers on all $m(a)$ scalars; and retain geometry-diverse blocks passing balance, overlap, and outcome-variation checks. These are matched observational evidence unless geometry was randomized by intervention.

b) *Gates.*: Hard thresholds keep the suite from becoming a soft narrative: G1: confounded amp $\text{AUROC} \geq 0.85$, paired amp ≤ 0.60 ; G2: perturbation-trained paired value ≤ 0.65 ; G3: paired-pert $\text{AUROC} \geq 0.10$; G4: closed-loop gap ≥ 0.15 against the amplitude heuristic. Failing G1–G2 means the labels are contaminated; failing G3–G4 means the “fix” did not change decisions. The cloneable artifact (victim_baseline/) prints this checklist from JSON so reviewers can reproduce the same pass/fail bits.

IV. EXPERIMENTS

a) *Setup.*: MuJoCo FactorTask with layout-disjoint splits under victim_baseline/. Confounded data replay a fixed expert on a fixed layout with joint noise $\sigma \in \{0.02, 0.04, 0.06, 0.08\}$; paired data hold amplitude fixed and

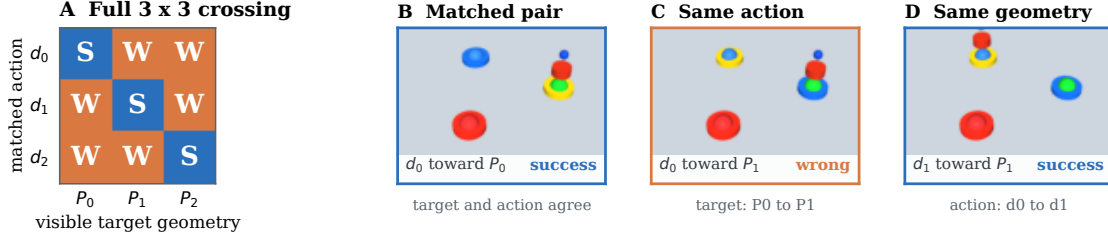


Fig. 3. The complete action $\times$ geometry crossing from actual MuJoCo rollouts. Each row holds the candidate action fixed; each column changes only the visible target geometry. The diagonal is the matched-direction success pattern; off-diagonal cells are non-success outcomes.

Algorithm 1 AGCD (summary)

- 1: Collect $\mathcal{D}_c$ (expert+noise on fixed geometry)
 - 2: Construct candidate set A with matched scalar summaries $m(a)$
 - 3: **for** each layout group G **do**
 - 4: execute the complete crossing $A \times G$ and record true rollout outcomes
 - 5: reject incomplete, scalar-mismatched, or outcome-constant groups
 - 6: **end for**
 - 7: Split $\mathcal{D}_p$ by layout before expanding candidate rows
 - 8: **G1:** AUROC($s \rightarrow y$) high on $\mathcal{D}_c$, ≈ 0.5 on $\mathcal{D}_p$
 - 9: Train amp head on $\mathcal{D}_c$; eval on $\mathcal{D}_p$ **G2:** collapse
 - 10: Train identical MiniWAM $_c$ / MiniWAM $_p$; evaluate paired ranking
 - 11: Run closed-loop $\arg \max_a f(a, g)$ on unseen layouts
 - 12: **G3–G4:** paired recovers AUROC and closed-loop gap ≥ 0.15
-

randomize seat geometry across three matched directions. MiniWAM optimizes $\text{BCE}(\text{success}) + \ell_2(\text{future})$; the FACT-style head adds a temporal progress target. Architecture is fixed within each comparison—only the collection recipe changes. Questions: (Q1) do labels leak through amplitude? (Q2) does a WAM-style head fall and recover? (Q3) does the gap transfer across tasks, engines, and vision? (Q4) are failures toxic, or only unmatched amplitude?

b) Tasks and outcomes.: The controlled suite spans reach, push, pick-place, and insert. Reach isolates continuous endpoint geometry; push adds object displacement; pick-place adds grasp and release; insert makes target occupancy decisive. Binary tables use task success. The dynamic carry-release control additionally preserves the three physical outcomes {success, wrong-target, drop}, preventing all failure modes from collapsing into one label. Labels always come from the rollout’s terminal geometry/contact state, never from candidate name, action magnitude, or a hand-written direction-equals-target rule.

c) Models and observations.: The amplitude heuristic uses only $m(a)$. Action-only receives direction and chunk statistics; geometry-only receives target/object layout; the joint model receives both. MiniWAM predicts terminal value

plus a short future state. The FACT-style Transformer predicts temporal progress, terminal value, and the same short future. RGB-centroid is a controlled perception upper bound; frozen DINOv2 [33] tests whether generic visual features retain the recipe contrast. All comparisons freeze the candidate generator and change only the training recipe.

d) Statistical unit.: Candidate rows from one layout are dependent, so splits and resampling use the layout (or episode) as the atomic unit. Main rates are means over five seeds; confidence intervals are nonparametric layout-level bootstrap intervals. We report the paired difference $\Delta = \text{metric}_{\text{joint}} - \text{metric}_{\text{action}}$ directly, rather than inferring significance from overlap between two marginal intervals. Multi-class results use one-vs-rest macro-AUROC so rare drop/wrong-target classes cannot disappear inside accuracy.

A. Q1: Does the shortcut exist in the labels?

Under the RL Bench-style victim (expert + $\sigma \in \{0.02, 0.04, 0.06, 0.08\}$), failed moves are $\sim 3\times$ larger in mean L2 than successes (0.102 vs 0.034; bootstrap 95% CI for mean-L2 AUROC [0.862, 0.939]; Table II). Fig. 4 makes the mechanism explicit: as σ rises, success collapses from 1.00 to 0.14 while mean action L2 grows monotonically (pooled amp AUROC 0.914). Amplitude alone is a strong success predictor; transfer to paired layouts collapses to chance. Paired collection also shifts failures from pure drops to *wrong-target* outcomes—geometry decides y once magnitude is controlled.

B. Q2: Does a WAM-style head fall for it—and recover?

Table III is the decision-level contrast: same layouts, same closed-loop rule, only the collection recipe changes. Figs. 5–6 separate ranking quality from its decision consequence. Under a shared large- n protocol (5 seeds, $n=120$), MiniWAM seats / RGB / frozen DINOv2 and a FACT-style progress/value Transformer (2.3M) all show the *same qualitative* pattern—pert. CL ≈ 0.33 , paired CL ≈ 0.95 –0.97, gap ≈ 0.62 –0.65—with seed-level scatter that is not a copied row. Capacity does not erase the leak; the FACT-style head does not claim parity with FACT’s released video WAM. The collapse is also not a logistic artifact: a scaled joint head on paired labels still only reaches 0.389.

Policy improvement. We next freeze each evaluator, score the same matched candidate bank on a layout-disjoint

TABLE II
SHORTCUT AUDIT AND ITS MECHANISM. CONFOUNDED ROWS USE $n=250$; PAIRED ROWS USE $n=120$ LAYOUTS. THE SAME ACTION AMPLITUDE IS UNINFORMATIVE ONLY AFTER THE GEOMETRY CROSSING IS ENFORCED.

Check	Confounded	Paired	Interpretation
Mean L2 $\rightarrow$ success	0.903	0.501	scalar shortcut disappears under amplitude control
Noise $\sigma \rightarrow$ success	0.891	—	larger perturbations are more likely to fail
Amp head (train on confounded)	0.907	0.500 [†]	in-domain score does not transfer
Action+geometry (train on paired)	—	0.765	geometry becomes usable after relabeling

[†] evaluated on the paired holdout.

TABLE III
DUAL-RECIPE DECISION AND POLICY-UPDATE LOOPS. DIRECT ROWS USE ARGMAX OVER SCALAR-MATCHED DIRECTIONS ($n=120$ LAYOUTS/SEED). POLICY ROWS DISTILL FEEDBACK INTO IDENTICAL POLICIES ON A DISJOINT UPDATE SPLIT AND EXECUTE WITHOUT THE EVALUATOR (n AS LISTED). ENTRIES ARE MEAN $\pm$ STD OVER FIVE SEEDS.

Family	Observation	Pert CL $\downarrow$	Paired CL $\uparrow$	Gap $\uparrow$	Runs
MiniWAM	privileged seats	0.328 $\pm$ 0.035	0.973$\pm$0.009	0.645$\pm$0.036	$n=120\times 5$
MiniWAM	RGB centroids	0.328 $\pm$ 0.035	0.948$\pm$0.041	0.620$\pm$0.066	$n=120\times 5$
MiniWAM	frozen DINOv2	0.328 $\pm$ 0.035	0.947$\pm$0.015	0.618$\pm$0.044	$n=120\times 5$
FACT-style WAM (2.3M)	privileged seats	0.330 $\pm$ 0.032	0.973$\pm$0.009	0.643$\pm$0.033	$n=120\times 5$
Evaluator $\rightarrow$ policy	privileged; evaluator-free test	0.296 $\pm$ 0.046	0.976$\pm$0.012	0.680$\pm$0.040	$n=90\times 5$
Evaluator $\rightarrow$ policy	frozen DINOv2; BC init.; RGB test	0.565 $\pm$ 0.136	0.925$\pm$0.017	0.360$\pm$0.125	$n=120\times 5$
Evaluator $\rightarrow$ policy	dynamic push	0.271 $\pm$ 0.094	0.993$\pm$0.010	0.722$\pm$0.101	$n=90\times 5$
Evaluator $\rightarrow$ policy	dynamic carry-release	0.349 $\pm$ 0.032	1.000$\pm$0.000	0.651$\pm$0.032	$n=90\times 5$

Within each comparison, architecture, update rule, and held-out candidates are fixed; training cardinality is reported and stressed separately below.

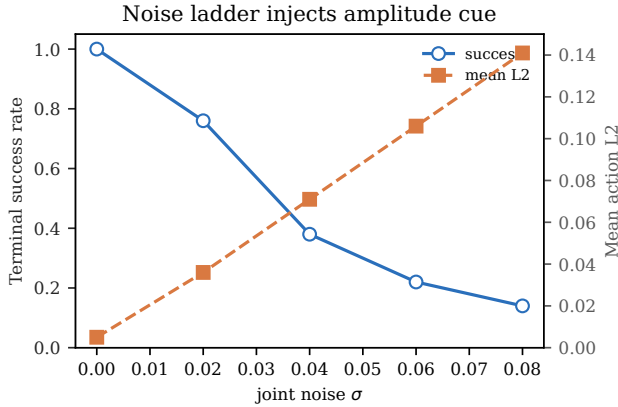


Fig. 4. Expert-perturbation CF couples failure with amplitude.

update split, and distill its temperature-scaled preferences into identical geometry-conditioned policies. At test time the evaluator is absent. For the RGB study, evaluator, eight-layout BC initialization, feedback update, and test layouts are mutually disjoint; all updates use the same KL=0.1 anchor to the BC policy. We selected the smallest competent BC budget on development seed 0; all reported RGB statistics use untouched seeds 1–5. Across five seeds and 450 held-out layouts, no update reaches 0.340 ± 0.052 , confounded feedback reaches 0.296 ± 0.046 , paired feedback reaches 0.976 ± 0.012 , and oracle feedback reaches 0.969 ± 0.020 . Paired minus confounded is 0.680 ± 0.040

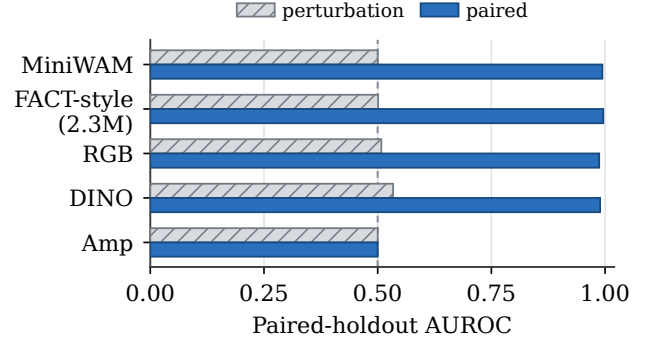


Fig. 5. **Paired-holdout ranking.** With the same heads, paired-geometry training recovers near-perfect AUROC while perturbation training collapses.

(seed-bootstrap 95% CI $[0.644, 0.704]$; 5/5 positive). Thus confounded feedback does not improve the proposal policy, whereas paired feedback transfers the recovered geometry dependence into its parameters rather than relying on test-time reranking. The same test is stronger under contact. Velocity-controlled push with randomized mass/friction gives 0.271 ± 0.094 vs 0.993 ± 0.010 (pooled layout-bootstrap gain CI $[0.680, 0.764]$). Carry-release gives 0.349 ± 0.032 vs 1.000 ± 0.000 (CI $[0.607, 0.696]$). Its confounded policy learns the partial lesson “do not release early”—drop falls to zero—but 65.1% of executions still reach a wrong target. Paired feedback supplies the missing action–target interaction, converting that semantic failure into correct

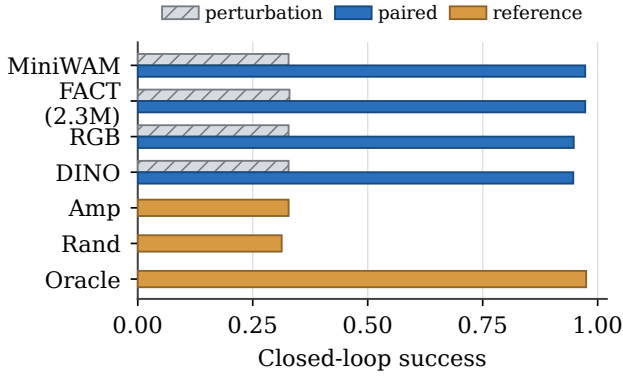


Fig. 6. **Closed-loop selection.** Paired-geometry training approaches the oracle; perturbation training collapses toward amplitude/random references.

placement (5/5 seeds). Finally, replacing coordinates with frozen DINOv2 RGB features and initializing all policies from the same eight-layout BC split gives no-update success 0.645 ± 0.154 , confounded-feedback success 0.565 ± 0.136 , and paired-feedback success 0.925 ± 0.017 on clean held-out images. Paired feedback therefore improves an already competent policy by 0.280 (pooled layout-bootstrap CI [0.245, 0.317]), whereas the confounded update is worse than no update. Identical augmentation during policy distillation retains paired success 0.882 under Gaussian noise, 0.955 under blur, 0.838 under rectangular occlusion, 0.923 under crop, and 0.905 under channel permutation; the paired-minus-no-update CI is positive in every condition and all five seeds are positive. The evaluator is absent in every reported execution.

Table IV fills every transfer cell: confounded training looks perfect in-domain and collapses under amplitude control; paired MiniWAM is the only recipe that both ranks and decides correctly. A scaled logistic joint head trained on paired data reaches CL only 0.389—nonlinear fusion helps, but protocol dominates. Across five seeds the MiniWAM gap is 0.65 ± 0.10 (paired AUROC 0.993 ± 0.006 , CL 0.950 ± 0.036).

C. Controls: scale, capacity, and sample size

Table IV compresses the non-story: larger splits, alternate CF (joint noise or gripper delay), and a width sweep all preserve the same qualitative gap—consistent with a *label* confound, not underfitting or a tiny-split artifact. Fig. 7 shows paired recovery is cheap: gap 0.50 already at 28 layouts, plateauing near 0.69. Keeping two of three matched directions per training layout cuts rollout cells by 33%, while evaluation uses the full grid. At 28/56/112 layouts, sparse CL is 0.886/0.989/1.000 versus 0.979/1.000/1.000 full (five seeds): cost is visible only at the lowest data regime. With 5/7 directions, random CL falls to 0.24/0.29 while oracle CL stays 0.94/0.96.

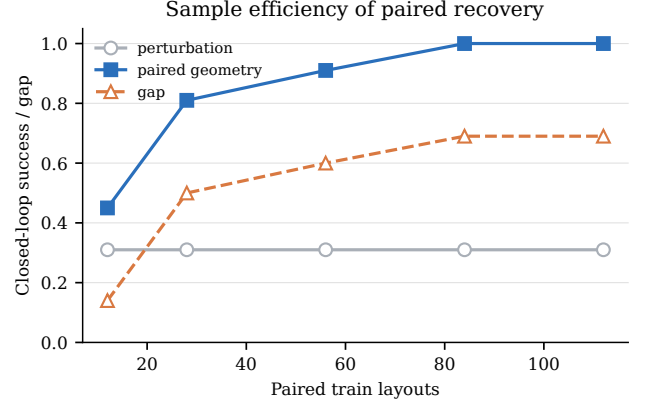


Fig. 7. Paired recovery is data-efficient (gap vs. #paired layouts).

D. Q3: Transfer—tasks, engines, vision

Fig. 8a shows perturbation CL at 0.31–0.42 versus paired CL 1.00 on four tasks. Removing label leakage does not solve visual optimization: paired/perturbation AUROC is 0.533/0.487 (CNN), 0.493/0.488 (ResNet18), and 0.986/0.550 (frozen DINOv2). MuJoCo [34], PyBullet [35], and SAPIEN [36] match by design under the shared kinematic contract (gap 0.620), excluding an API artifact rather than proving physics transfer. Cross-task evaluation without task IDs preserves paired CL 1.00. In Fig. 8b, geometry noise produces a smooth $1.00 \rightarrow 0.75$ decline, while $\approx 5\%$ paired data saturates the mixture curve. Continuous target jitter plus 6–10 mm endpoint noise retains a 0.37–0.48 joint-minus-action macro-AUROC gain (600 rows/engine).

For contact-rich controls (Fig. 8c), randomized mass/friction push gains 0.481 (CI [0.411, 0.568]); three-outcome carry-release gains 0.207 ([0.171, 0.236]). Every seed is positive. These are predictor metrics; Table III tests evaluator-free policy updates.

E. External real-video label audit

We tested whether a bounded BridgeData V2 [16] LeRobot shard could support an external outcome audit. It cannot: no common success label is present. Before fitting any model, two blinded raters judged 120 shared clips drawn from 26 task strings and agreed perfectly ($\kappa=1.00$). An endpoint-progress proxy agreed with that rater consensus on only 52.5% (95% CI [43.3, 61.7]%; balanced accuracy 41.3%); 113/120 clips showed visible progress. We therefore reject the proxy and withhold all public-data outcome/amplitude AUROC claims. The shard, matching code, and negative audit remain in the artifact. BridgeData, DROID [21], or any expert-heavy collection must establish verified outcome variation before a failure-shortcut audit is interpretable.

F. Q4: Are failure data toxic—or unmatched amplitude?

Table V separates toxicity from amplitude matching. FACT-style matched-amplitude policy failures stay near chance (AUROC 0.54); mixing larger failed moves with

TABLE IV
TRANSFER AND CAPACITY CONTROLS. THE SAME EVALUATION LAYOUTS ARE USED THROUGHOUT; ONLY THE TRAINING RECIPE, FEATURE SOURCE, OR MODEL WIDTH CHANGES.

Train \ Eval	(a) Full train $\times$ eval transfer			(b) Controls that preserve the gap			
	Conf. AUROC	Paired AUROC	Paired CL	Control	Conf. amp	Pert. CL	Paired CL / Gap
Amp $\leftarrow$ conf.	0.884	0.486	0.222	Scale-up (150/240)	0.909	0.347	0.972 / 0.625
Scaled joint $\leftarrow$ paired	0.512	0.522	0.389	Joint-noise CF	0.896	0.271	0.979 / 0.708
MiniWAM $\leftarrow$ conf.	1.000	0.500	0.250	Gripper-delay CF	0.931	0.271	0.979 / 0.708
MiniWAM $\leftarrow$ paired	0.381	0.983	0.972	Width 64 $\rightarrow$ 512	—	0.313	1.000 / 0.688
Oracle / Random / Amp-heur.	—	—	1.00 / 0.33 / 0.22				

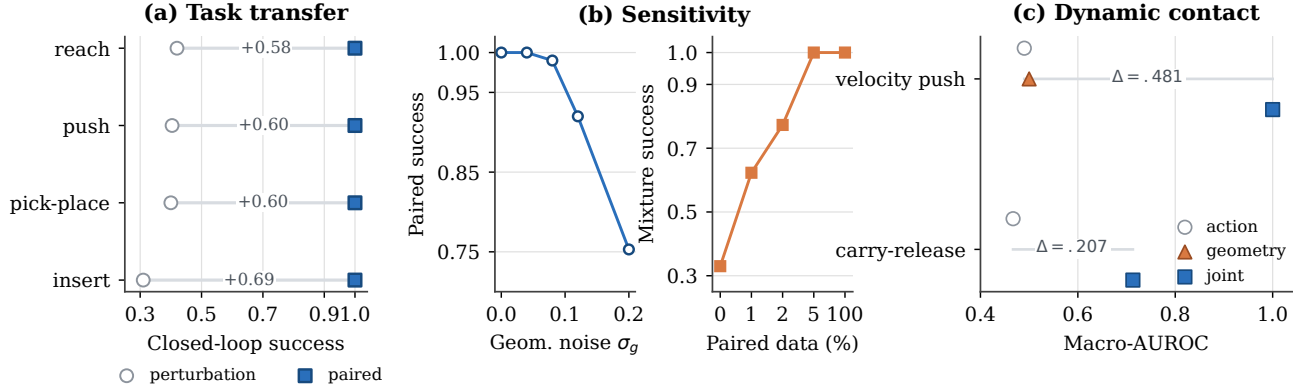


Fig. 8. **Transfer and robustness.** (a) Recipe gaps across tasks. (b) Geometry-noise and paired-data sensitivity. (c) Dynamic-contact AUROC; Δ is joint minus action (95% CI). Five-seed means.



Fig. 9. **Representative real-tabletop executions.** Top: successful red-cube pick-and-place. Bottom: placement failure. This is qualitative physical evidence; quantitative paired-intervention results remain simulation based.

expert successes recreates AUROC 1.00. Frozen PointWorld and FACT value probes on our synthetic observations stay near chance (≈ 0.51). We did not train public weights on the victim chain; AGCD targets the *wiring*, not an accusation that every release already leaks.

V. DISCUSSION

a) Protocol before loss.: A regularizer can suppress a representation of a confound that remains in the labels. Paired geometry removes that cue at collection time, so AGCD is upstream of representation learning: it gates failure-driven policy updates by requiring (i) recipe amplitude AUROC, (ii)

TABLE V
FAILURE DATA ARE NOT INTRINSICALLY TOXIC—UNMATCHED AMPLITUDE IS.

Recipe	Amp $\rightarrow$ success AUROC
Expert-perturbation CF (victim)	0.90
FACT-style matched-amp failures	0.54
Same fails mixed with larger moves	1.00
Frozen FACT / PointWorld on our obs	≈ 0.51

amplitude-controlled paired AUROC, and (iii) closed-loop selection against an amplitude heuristic. Failure data are useful only after this ranking test; the frozen FACT/PointWorld probes and the public-video audit are controls, not claims that every released WAM is flawed. This distinction matters for self-improvement: a failure trajectory can teach a world model about consequences while still being an invalid target for an outcome head if its label is predictable from perturbation size.

b) Decision consequence.: The paired protocol is therefore a data-to-decision gate, not a new backbone or loss. Once the joint head passes the crossing test, its score can be used for candidate ranking, rejection, or evaluator-guided policy updates. If it fails, additional capacity or calibration cannot recover the missing action–geometry signal; the collection recipe must change first.

c) *Why calibration is downstream.*: Temperature scaling preserves candidate ordering, while conformal rejection can reduce coverage without removing a shortcut from accepted actions. Neither creates geometry dependence when amplitude already reveals the label. AGCD therefore tests paired ranking and execution before calibrating selective risk; the reverse order can yield calibrated scores with physically wrong choices.

d) *Cost and scaling.*: A complete $m \times n$ crossing is deliberately diagnostic rather than a proposal to densely tile a continuous action space. Its purpose is to expose identifiability before expensive training. Sparse matched directions reduce collection by one third with little loss except at the smallest regime (Sec. IV-C). In higher dimensions, stratified matching and calipers reduce cost, but change the claim from randomized counterfactual evidence to observational evidence.

e) *Scope and reproducibility.*: The main evidence is the MuJoCo FactorTask, with dynamic-contact and cross-engine stress tests. RGB/DINO inputs use controlled visual markers, and the BridgeData result is deliberately a negative proxy audit; neither implies sim-to-real transfer. The release contains the factor generator, JSON gate reports, model/task scripts, and tests; no checkpoint is needed to reproduce the diagnostic. Fig. 9 documents the physical setup and outcome contrast; exact paired interventions on hardware remain the next validation step and do not change the paper’s primary claim.

f) *Physical validation protocol.*: The hardware study will preserve the diagnostic contrast rather than report isolated successful grasps: scalar-matched candidates are re-played across object layouts, Action-only and joint scores rank the same candidates, and outcomes record grasp success, wrong-object/contact events, and rejection. This small-scale test asks whether visible geometry changes real decisions; it does not serve as a broad manipulation benchmark. Controlled tabletop resets support the strongest comparison; if reset error cannot be bounded, results will be reported as matched trials rather than exact counterfactuals.

VI. CONCLUSION

Expert perturbations can inject an action-amplitude shortcut into outcome heads. AGCD audits that recipe, holds action scalars fixed while varying geometry, retrains the same head, and tests both selection and policy improvement. The confounded predictor fails both; paired feedback restores ranking and transfers to an evaluator-free policy. AGCD is a falsifiable data-to-decision-to-learning contract for failure-driven WAMs, not a new backbone.

REFERENCES

- [1] D. Ha and J. Schmidhuber, “Recurrent world models facilitate policy evolution,” *NeurIPS*, 2018.
- [2] D. Hafner et al., “Mastering diverse control tasks through world models,” *Nature*, 2025.
- [3] Y. Shi et al., “Overcoming statistical bias in action-controllable world models,” arXiv:2608.04653, 2026.
- [4] Q. Peng et al., “FACT: Failure-aware causal training for world-action models,” arXiv:2608.10232, 2026.
- [5] H. Bi et al., “Motus2: A self-evolving general world model for dexterous manipulation,” arXiv:2608.30237, 2026.
- [6] G. Zhao et al., “FLARE: A failure-aware framework for autonomous correction and recovery in visual-language robotic manipulation,” arXiv:2608.26645, 2026.
- [7] S. James et al., “RLBench: The robot learning benchmark & learning environment,” *IEEE RA-L*, 2020.
- [8] Y. Zhao et al., “GWM-VLA: Geometry-aware latent world modeling for vision-language-action learning,” arXiv:2608.07619, 2026.
- [9] X. Sun et al., “VGGT-World: Transforming VGGT into an autoregressive geometry world model,” arXiv:2603.12655, 2026.
- [10] J. Wang et al., “VGGT-Ω,” arXiv:2605.15195, 2026.
- [11] Y. Hu et al., “VGGT4D: Mining Motion Cues in Visual Geometry Transformers for 4D Scene Reconstruction,” arXiv:2511.19971, 2025.
- [12] H. Arai et al., “ACT-Bench: Towards action controllable world models for autonomous driving,” arXiv:2412.05337, 2024.
- [13] C. Oefinger et al., “Validate the dream before you trust its verdict,” arXiv:2607.07196, 2026.
- [14] R. Geirhos et al., “Shortcut learning in deep neural networks,” *Nature Machine Intelligence*, 2020.
- [15] M. Arjovsky et al., “Invariant risk minimization,” arXiv:1907.02893, 2019.
- [16] H. Walke et al., “BridgeData V2: A Dataset for Robot Learning at Scale,” *CoRL*, 2023.
- [17] W. Huang, Y.-W. Chao, A. Mousavian, M.-Y. Liu, D. Fox, K. Mo, and L. Fei-Fei, “PointWorld: Scaling 3D world models for in-the-wild robotic manipulation,” arXiv:2601.03782, 2026.
- [18] C. Chi, Z. Xu, S. Feng, E. Cousineau, Y. Du, B. Burchfiel, R. Tedrake, and S. Song, “Diffusion Policy: Visuomotor policy learning via action diffusion,” *RSS*, 2023.
- [19] S. Ross, G. J. Gordon, and J. A. Bagnell, “A reduction of imitation learning and structured prediction to no-regret online learning,” *AISTATS*, 2011.
- [20] P. de Haan, D. Jayaraman, and S. Levine, “Causal confusion in imitation learning,” *NeurIPS*, 2019.
- [21] A. Khazatsky et al., “DROID: A large-scale in-the-wild robot manipulation dataset,” *RSS*, 2024.
- [22] Open X-Embodiment Collaboration et al., “Open X-Embodiment: Robotic learning datasets and RT-X models,” *ICRA*, 2024.
- [23] C. Finn, I. Goodfellow, and S. Levine, “Unsupervised learning for physical interaction through video prediction,” *NeurIPS*, 2016.
- [24] C. Finn and S. Levine, “Deep visual foresight for planning robot motion,” *ICRA*, 2017.
- [25] D. Hafner, T. Lillicrap, I. Fischer, R. Villegas, D. Ha, H. Lee, and J. Davidson, “Learning latent dynamics for planning from pixels,” *ICML*, 2019.
- [26] S. Dasari et al., “RoboNet: Large-scale multi-robot learning,” *CoRL*, 2020.
- [27] A. Mandlekar et al., “What matters in learning from offline human demonstrations for robot manipulation,” *CoRL*, 2022.
- [28] B. Liu, Y. Zhu, C. Gao, Y. Feng, Q. Liu, Y. Zhu, and P. Stone, “LIBERO: Benchmarking knowledge transfer for lifelong robot learning,” *NeurIPS Datasets and Benchmarks*, 2023.
- [29] O. Mees, L. Hermann, E. Rosete-Beas, and W. Burgard, “CALVIN: A benchmark for language-conditioned policy learning for long-horizon robot manipulation tasks,” *IEEE RA-L*, 2022.
- [30] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” *ICML*, 2017.
- [31] Y. Geifman and R. El-Yaniv, “SelectiveNet: A deep neural network with an integrated reject option,” *ICML*, 2019.
- [32] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster, “Conformal risk control,” *ICLR*, 2024.
- [33] M. Oquab et al., “DINOv2: Learning robust visual features without supervision,” *TMLR*, 2024.
- [34] E. Todorov, T. Erez, and Y. Tassa, “MuJoCo: A physics engine for model-based control,” *IROS*, 2012.
- [35] E. Coumans and Y. Bai, “PyBullet, a Python module for physics simulation for games, robotics and machine learning,” 2016–2021. [Online]. Available: <https://pybullet.org>
- [36] F. Xiang et al., “SAPIEN: A Simulated Part-Based Interactive Environment,” *CVPR*, 2020.